

Mathematics

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Chapter 1

Algebra

1.1 Groups

1.1.1 Magmas and monoids

Magmas and properties

Definition 1.1. A magma is a pair $(M, *)$ with M a set and

$$* : M \times M \rightarrow M$$

an internal composition law.

Definition 1.2. A magma $(M, *)$ is said to be associative if $\forall a, b, c \in M, (a*b)*c = a*(b*c)$.

Definition 1.3. A magma $(M, *)$ is said to be commutative if $\forall a, b \in M, a*b = b*a$.

Definition 1.4. An identity element of a given magma $(M, *)$ is an element $e \in M$, such that $\forall a \in M, e*a = a*e = a$.

Theorem 1.5. *If a magma $(M, *)$ has an identity element, then it is unique.*

Proof. If $e, e' \in M$ are identity elements for $*$, then $e = e*e' = e'$. □

Definition 1.6. Given a magma $(M, *)$ with an identity element $e \in M$. An element $a \in M$ is said to be invertible if there exists $b \in M$ such that $a*b = b*a = e$. b is then called an inverse of a .

Products of magmas

Definition 1.7. Given two magmas $(M, *)$ and $(N, \bullet)$, we define a canonical structure of magma on the set $M \times N$, defined by the following internal composition law:

$$((a, b), (a', b')) \mapsto (a*a', b \bullet b')$$

Proposition 1.8. *Let M and N be two magmas.*

- (i) *If M and N are associative (resp. commutative), then $M \times N$ is associative (resp. commutative),*
- (ii) *If M and N have identity elements, then $M \times N$ has an identity element.*

Proof. Admitted. □

Proposition 1.9. *Let M and N be two magmas with identity elements. If $a \in M$ and $b \in N$ are invertible, then $(a, b) \in M \times N$ is invertible.*

Proof. Admitted. □

Remark 1.10. The previous construction can be extended to any product of magmas. The propositions (1.8) and (1.9) then remain true.

Morphism of magmas

Definition 1.11. A morphism of magmas between $(M, *)$ and $(N, \bullet)$ is a map $f : M \rightarrow N$ such that

$$\forall a, b \in M, f(a * b) = f(a) \bullet f(b)$$

Proposition 1.12. *A composition of morphisms of magmas is a morphism of magmas.*

Proof. Admitted. □

Monoids and properties

Definition 1.13. A monoid is an associative magma with an identity element.

Theorem 1.14. *If $(M, *)$ is a monoid, then every invertible element $a \in M$ has a unique inverse, denoted by a^{-1} .*

Proof. Let $b, b' \in M$ be two inverses of a . Then we have

$$b = b * (a * b') = (b * a) * b' = b'.$$

□

Notation 1.15. If $(M, *)$ is a monoid, then for all $a \in M$, we denote:

$$a^0 = e \quad \text{and} \quad a^n = \underbrace{a * \cdots * a}_{n \text{ times}} \quad (\text{for } n \geq 1).$$

Proposition 1.16. *Given a monoid $(M, *)$, and $a, x, y \in M$,*

- (i) *If $a * x = a * y$ and a is invertible, then $x = y$,*

(ii) If x is invertible, then x^{-1} is invertible and $(x^{-1})^{-1} = x$.

(iii) If x and y are invertible, then $x * y$ is also invertible and $(x * y)^{-1} = y^{-1} * x^{-1}$,

(iv) If x is invertible, then for all $n \in \mathbb{N}$, x^n is also invertible and $(x^n)^{-1} = (x^{-1})^n$. Hence we can denote $x^{-n} = (x^n)^{-1}$,

Proof. (i) $x = (a^{-1} * a) * x = a^{-1} * (a * x) = a^{-1} * (a * y) = (a^{-1} * a) * y = y$.

(ii) Direct consequence of the invertible element definition.

(iii) $(x * y) * (y^{-1} * x^{-1}) = x * y * y^{-1} * x^{-1} = e$, and similarly for the other way.

(iv) It suffices to combine the two preceding properties in a proof by induction. □

Proposition 1.17. *Any product of monoids is a monoid.*

Proof. It is induced by (1.8). □

Definition 1.18. A morphism of monoids is a morphism between underlying magmas, such that it preserves the identity element.

1.1.2 Groups

Groups and subgroups

Definition 1.19. A group is a monoid such that every element is invertible.

Proposition 1.20. *Any product of groups is a group.*

Proof. It is induced by (1.17) and (1.9). □

Definition 1.21. Let $(G, *)$ be a group. A subset $H \subseteq G$ is said to be a *subgroup* of G if:

(i) The identity element $e \in H$,

(ii) H is closed under the composition law, i.e., $\forall x, y \in H, x * y \in H$,

(iii) H is closed under taking inverses, i.e., $\forall x \in H, x^{-1} \in H$.

Proposition 1.22. *Let $(G, *)$ be a group. A non-empty subset $H \subseteq G$ is a subgroup of G if and only if:*

$$\forall x, y \in H, \quad x * y^{-1} \in H.$$

Proof. Suppose H is a subgroup of G . Let $x, y \in H$. By condition (iii) of Definition 1.21, $y^{-1} \in H$. Then, by condition (ii), $x * y^{-1} \in H$.

Conversely, suppose $H \neq \emptyset$ and satisfies the given criterion. Since H is non-empty, there exists some $x \in H$. Applying the criterion to x and x yields $x * x^{-1} = e \in H$, which satisfies condition (i). Now, for any $y \in H$, applying the criterion to e and y gives $e * y^{-1} = y^{-1} \in H$, satisfying condition (iii). Finally, for any $x, y \in H$, we know $y^{-1} \in H$, so applying the criterion to x and y^{-1} gives $x * (y^{-1})^{-1} = x * y \in H$, satisfying condition (ii). □

Proposition 1.23. *Given a group G , any intersection of subgroups of G is a subgroup of G .*

Proof. Admitted. □

Morphism of groups

Definition 1.24. A morphism of groups is a morphism of underlying magmas.

Proposition 1.25. *Let G, G' be two groups and $f : G \rightarrow G'$ be a morphism of groups. Then $f(e_G) = e_{G'}$ and for all $x \in G$, $f(x^{-1}) = f(x)^{-1}$.*

Proof. Admitted. □

Proposition 1.26. *A composition of morphisms of groups is a morphism of groups.*

Proof. Induced by (1.12). □

Theorem 1.27. *Let G, G' be two groups and $f : G \rightarrow G'$ be a morphism of groups.*

- (i) *If H is a subgroup of G , then $f(H)$ is a subgroup of G' ,*
- (ii) *If H' is a subgroup of G' , then $f^{-1}(H')$ is a subgroup of G .*

Proof. Admitted. □

Definition 1.28. The kernel of a morphism of groups $f : G \rightarrow G'$ is defined by

$$\text{Ker}(f) = \{a \in G \mid f(a) = e_{G'}\}$$

Proposition 1.29. *The kernel of a morphism of groups $f : G \rightarrow G'$ is a subgroup of G .*

Proof. Induced by (1.27)-(i) □

Definition 1.30. The image of a morphism of groups $f : G \rightarrow G'$ is defined by

$$\text{Im}(f) = \{b \in G' \mid \exists a \in G, f(a) = b\}$$

Proposition 1.31. *The image of a morphism of groups $f : G \rightarrow G'$ is a subgroup of G' .*

Proof. Induced by (1.27)-(ii) □